Financing Frictions: DiD, advanced topics

Adrien Matray
(Federal Reserve Bank of Atlanta, CEPR)

Outline

DiD: more advanced topics

General message

There is a tension between causal inference and “aggregate decomposition”

NB: by “aggregate decomposition” I still abstract from the **missing intercept** problem and assume **SUTVA** is not violated

- For how to test properly test for SUTVA, see lecture ”Disaggregate and Aggregation Invariance”

What you should be able to do by the end

- Diagnose who identifies β under HDFE: a special subsample (multi-bank firms), intensive margin only, even with valid identification
- Spot when explicit or implicit weights and granularity make firm-level, average, and aggregate estimates disagree
- Detect mean reversion (FE with unbalanced covariates, base-year exposure shares) and fix it with deep lags or non-mean-reverting covariates

Three problems, in order: HDFE & representativeness $\rightarrow$ weights & granularity $\rightarrow$ mean reversion; the aggregation-invariance fix is next lecture

Outline

The costs of HDFE

- Common now to "saturate the regression" with high-dimensionality fixed effects (HDFE)

- Pb: narrowing down the variation can produce estimates that are far from "aggregate" response

The costs of HDFE

- Common now to “saturate the regression” with high-dimensionality fixed effects (HDFE)

- Pb: narrowing down the variation can produce estimates that are far from “aggregate” response

  1. Cells with variations left are special: e.g., “within-firm estimator” in banking

  2. Identification is narrowed down to intensive margin: e.g., market-year fixed effects in trade

Borrower fixed effects in banking: how is β identified?

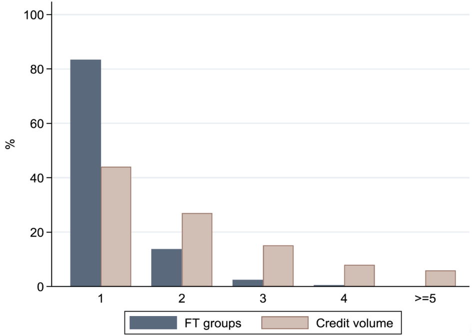
$$Log(credit_{i,b,t}) = \beta BankShock_{b,t} + \alpha_i \times \delta_t + \gamma_b$$

$\Rightarrow$ How is β identified?

Borrower fixed effects: identification rests on a small set of multi-bank firms

Usually a small fraction of firms borrow from multiple banks → how to deal with that?

Panel A: Firm-Time groups

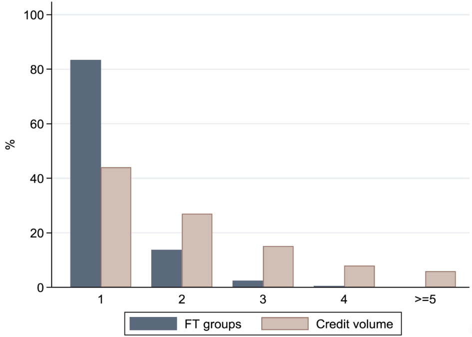


(Degryse, De Jonghe, Jakovljević, Mulier and Schepens (2019), *Journal of Financial Intermediation*)

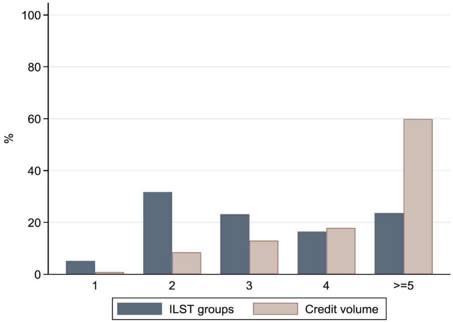
Borrower fixed effects: identification rests on a small set of multi-bank firms

Solution: create “fake firms”

Panel A: Firm-Time groups



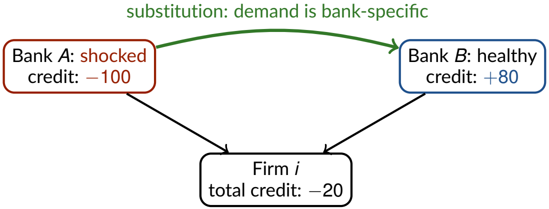
Panel B: Industry-Location-Size-Time groups



(Degryse, De Jonghe, Jakovljević, Mulier and Schepens (2019), *Journal of Financial Intermediation*)

Additional problems from high degree of disaggregation

1. Strong identifying assumption: loan demand is symmetric $\rightarrow$ potentially vastly overestimate the true magnitude

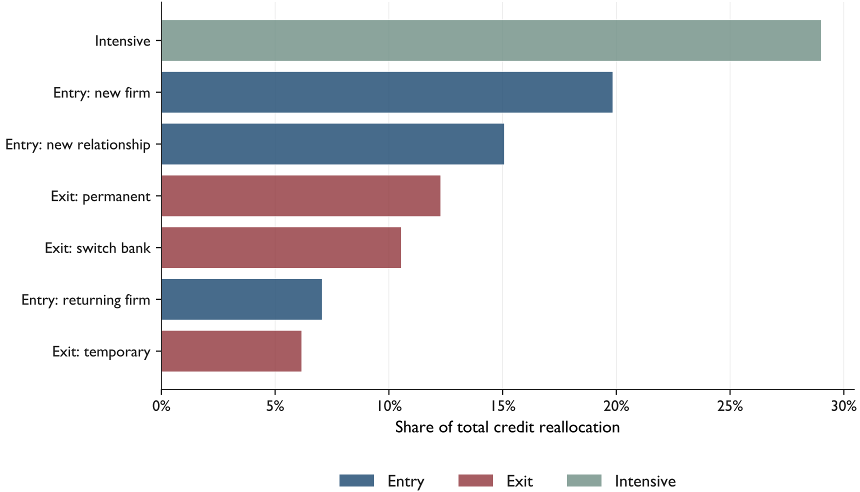


within-firm comparison (A vs B): -180 vs firm's true credit loss: -20

$\rightarrow$ assuming symmetric demand = assuming away the substitution arrow

Additional problems from high degree of disaggregation

2. LHS typically in log growth $\rightarrow$ captures only the intensive margin



French credit registry

Key takeway

Standard challenge with datasets that allow high level of disaggregation (e.g., credit registry, custom data):

- Incredibly powerful to control for various unobserved heterogeneity
  But comes at costs

Key takeway

Standard challenge with datasets that allow high level of disaggregation (e.g., credit registry, custom data):

- Incredibly powerful to control for various unobserved heterogeneity
  But comes at costs
  - Representativeness: subsamples where we still have variations with high-dimension FEs may have different elasticities than modal firm (Degryse, Jonghe, Jakovljevic, Mulier and Schepens, 2019)
  - Margin: within unit variation ⇒ typically focus on intensive margin and miss the extensive margin (Beaumont-Matray-Tu-Xu, 2026)
  - Substitution across lenders can drive the results ⇒ lender-firm effects may look very different from firm / bank / industry level effects ("semi-aggregate") (Pinardon-Touati, 2025)

Key takeway

Standard challenge with datasets that allow high level of disaggregation (e.g., credit registry, custom data):

- Incredibly powerful to control for various unobserved heterogeneity
  But comes at costs
  - Representativeness: subsamples where we still have variations with high-dimension FEs may have different elasticities than modal firm (Degryse, Jonghe, Jakovljevic, Mulier and Schepens, 2019)
  - Margin: within unit variation ⇒ typically focus on intensive margin and miss the extensive margin (Beaumont-Matray-Tu-Xu, 2026)
  - Substitution across lenders can drive the results ⇒ lender-firm effects may look very different from firm / bank / industry level effects (“semi-aggregate”) (Pinardon-Touati, 2025)
- Solution: vary level of aggregation of the data (e.g.: loan, bank, firm, geography)

→ But how you do it **matters a great deal!** (more on that later with Beaumont-Matray-Tu-Xu, 2026)

Outline

Challenge: explicit weighting

Common in applied macro to weight regressions to get “closer to (reduced-form) aggregate”
e.g.: commuting zone employment, total sales, total credit

$\Rightarrow$ Two problems

1. SUTVA rarely perfectly respected (some business stealing, input reallocation)

Challenge: explicit weighting

Common in applied macro to weight regressions to get "closer to (reduced-form) aggregate"
e.g.: commuting zone employment, total sales, total credit

⇒ Two problems

1. SUTVA rarely perfectly respected (some business stealing, input reallocation)

2. Extrem granularity of most datasets
   - Top 1% of firms = 75% total sales and 85% total investment in QFR data (Crouzet-Mehrotra, 2020)

Explicit weighting and granualirity

Extrem granuarity of most admin data has two implications:

1. “Average” elasticity reflects at best larger firms’ elasticity
   - Typically very different (e.g., less constrained)

Explicit weighting and granualirity

Extrem granuarity of most admin data has two implications:

1. “Average” elasticity reflects at best larger firms’ elasticity
   - Typically very different (e.g., less constrained)

2. Average elasticity reflects at worst idiosyncratic shocks on largest firms
   - Granularity $\rightarrow$ LLN no longer applies: $\mathcal{E}[\epsilon_{i,t}] \neq 0$
   - Idio shocks are important: firm-specific component explains around half of GDP growth (di Giovani et al., 14, 18, 23)

Illustrations

What better predicts firm growth: average firm growth at the industry level, or sum of sales industry growth?

Implicit weights have similar flavor

See next lecture Disaggregation and Aggregation Invariance

Outline

Mean reversion is potentially a big problem in DiD setting

- Most variables we study in CF mean revert and can have large and prolonged idiosyncratic shocks
  - The world we live in is less stable than we think!
- Problem 1: fixed effects with unbalanced covariates
- Problem 2: DiD with only aggregate shifters

Problem 1: fixed effects with unbalanced covariates

- Two types of firms: treated and control
  - Control larger than treated
  - You estimate
$$\Delta K_{i,t} = \beta \mathit{Treated} \times \mathit{Post} + \alpha_i + \delta_t$$
  - Worry? Solution?
  - Now let's assume $\Delta K_{i,t} = \rho \Delta K_{i,t-1}$

Treated firms will grow slower due to mean reversion

- The careful fix researchers use: control for ex-ante size differences
  - Size decile fixed once at a base year t_0 (before the dynamics), not updated each year
  - Ex-ante control, chosen to avoid the bad-control problem
- Equation to estimate:

$$\Delta K_{i,t} = \beta \,\text{Treated} \times \text{Post} + \text{Size decile}_{t_0} \times \text{Year} + FE$$

Monte carlo simulations (1,000)

- Two types of firms: “treated” and “control”
  - Control larger than treated
  - $\Delta K_{i,t} = \rho\, \Delta K_{i,t-1} + \epsilon_{i,t}$, with ρ = autocorrelation

    $\Rightarrow$ **By construction, the true treatment effect is zero**

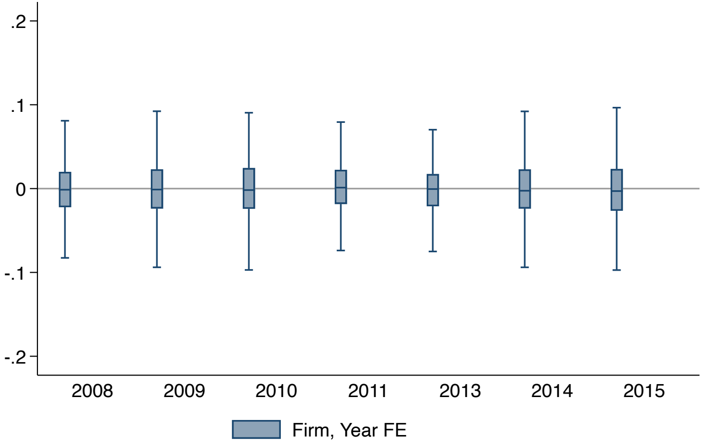
- Estimate:

$$\Delta K_{i,t} = \sum_{\substack{t \in [2009-2014] \\ t \neq 2012}} \beta_t (\text{Treated}_i \times 1_{Year=t}) + \alpha_i + \text{Size decile}_{t_0} \times \delta_t + \varepsilon_{it}$$

t_0 = fixed base year (ex ante); we vary $t_0 = 2009, \ldots, 2012$ across panels

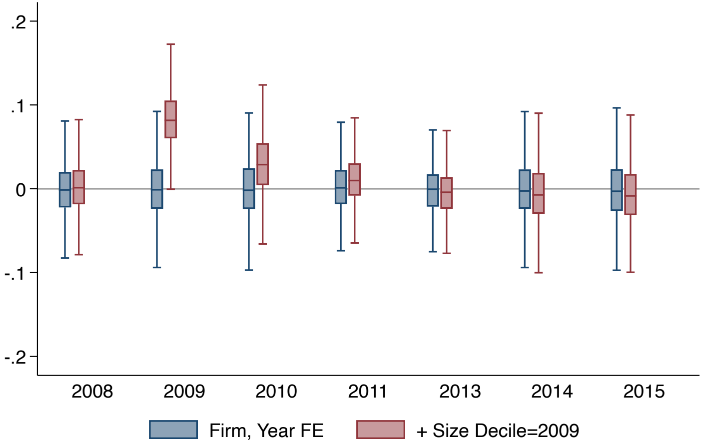
Plotting the β_t from the simulation

- With only firm and year FE: treatment effect correctly zero



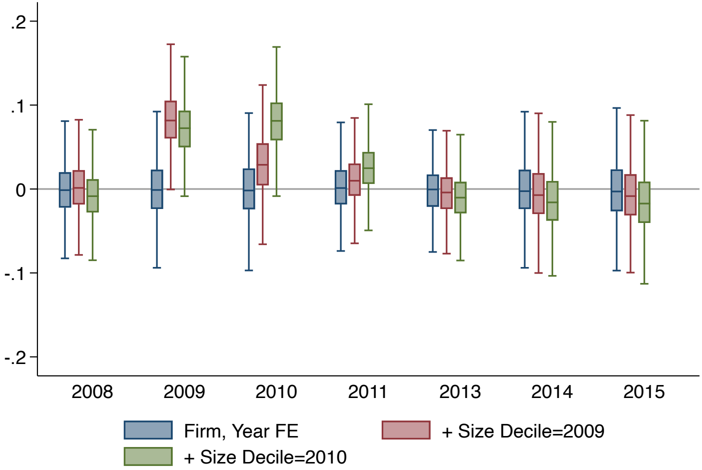
Plotting the β_t from the simulation

- Fix size decile at $t_0 = 2009$ (ex ante) $\rightarrow$ treated firms need to "jump," then mean revert



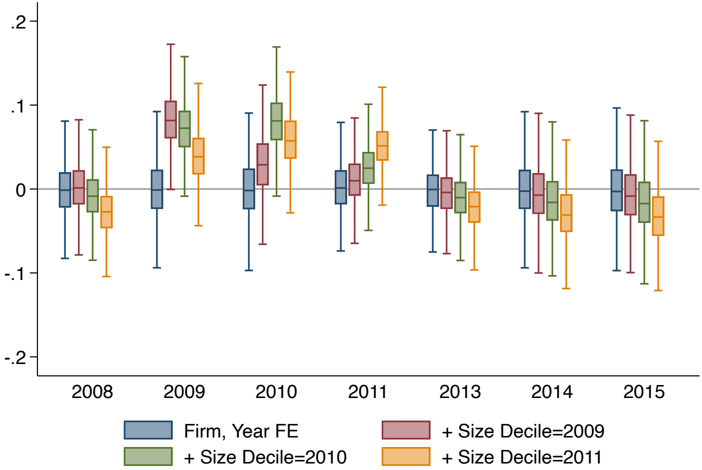
Plotting the β_t from the simulation

- Fix size decile at $t_0 = 2010$ (ex ante) $\rightarrow$ treated firms have a series of good draw up to 2010



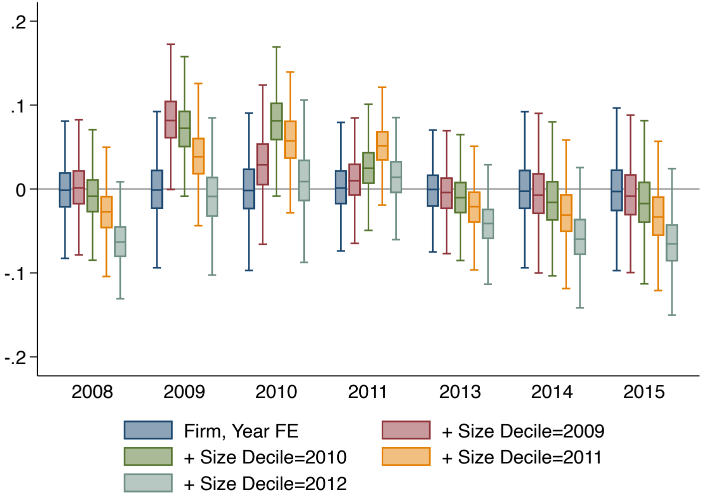
Plotting the β_t from the simulation

- Fix size decile at $t_0 = 2011$ (ex ante) $\rightarrow$ treated firms have a series of good draw up to 2011



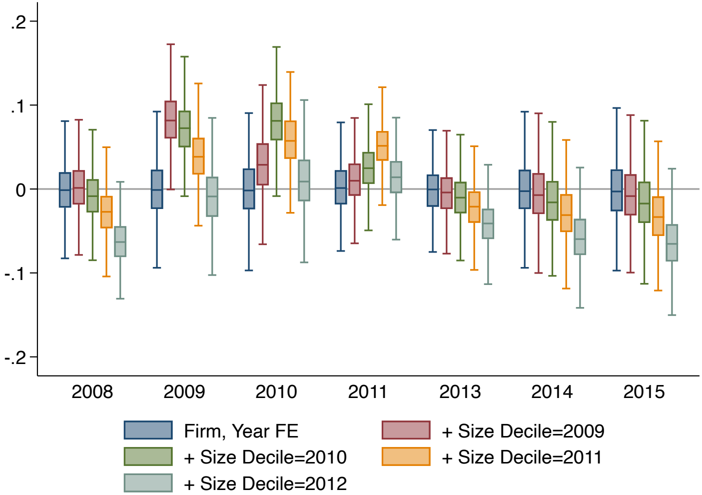
Plotting the β_t from the simulation

- Fix size decile at $t_0 = 2012$ (ex ante) $\rightarrow$ treated firms have a series of good draw up to 2012



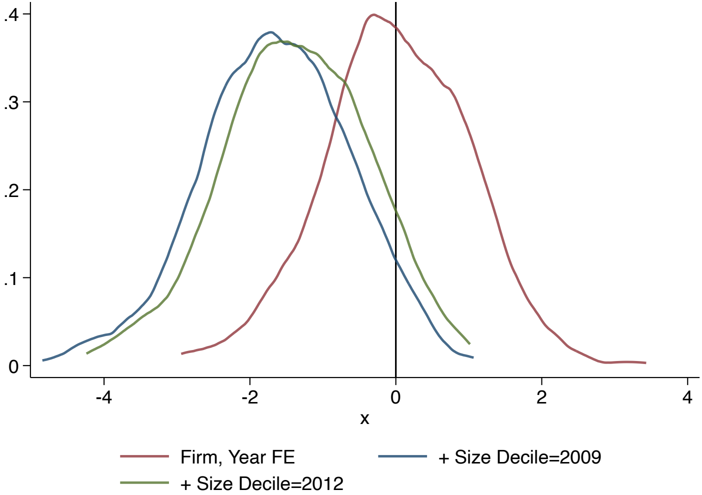
Plotting the β_t from the simulation

- Exact shape post sorting depends on exact process, but pattern always the same



Mean reversion produces average downward bias

- t-stat from the average DiD estimation: Adding the ex-ante size-decile control leads to significant negative results



Wait: the true effect is zero, so why is β_t negative?

- Treatment was set to zero by construction, yet the event study trends down after sorting on the ex-ante size decile

$\rightarrow$ What is the ex-ante size decile actually selecting on?

Mean reversion in firm size: intuition

- Control firms much bigger than treated firms
- Including the base-year (t_0) size decile "forces" a comparison between treated and control firms that were the same size in t_0 (ex ante) $\rightarrow$ which are they?
  - In top deciles: treated firms that got positive abnormal shock $\Rightarrow$ that's why they can be big enough
  - Bottom deciles: control firms that got negative abnormal shock
- Once you stop imposing similar size, firms will mean revert $\Rightarrow$ treated firms will decline relative to control firms

Is there a solution?

- Using average over whole period does not change the problem

- Two options
  - Use deep lag to give time to the variable to return to its steady state
  - Use variables that are less likely to mean revert: industry, location, age

Same risk if instead you cut your sample ex ante

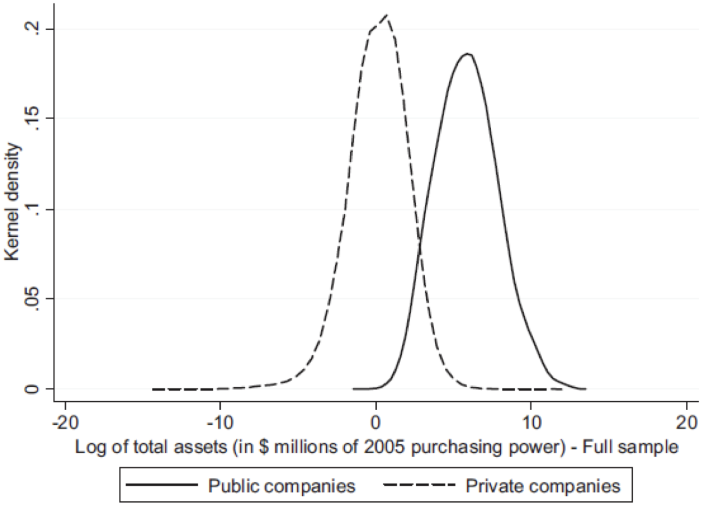
Debate differences in behaviors listed/private for investment

- Asker, J., Farre-Mensa, J., & Ljungqvist, A. (2015). Corporate Investment and Stock Market Listing: A Puzzle? *Review of Financial Studies*

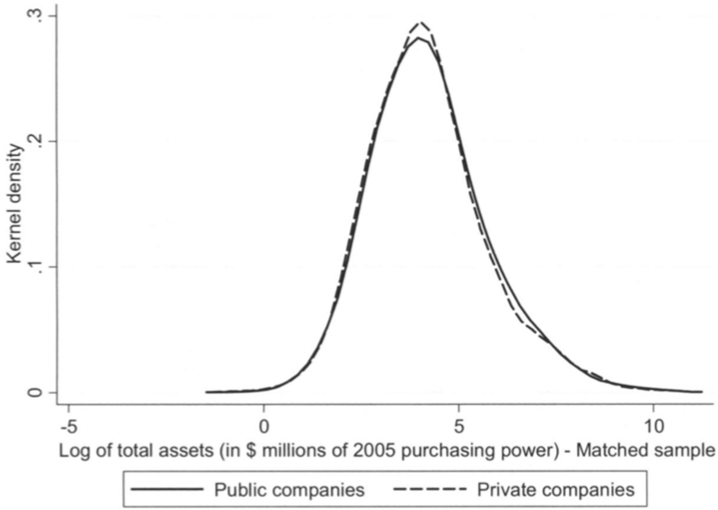
  VS

- Maskimovic, Phillips and Yang (2018) Do Public Firms Respond to Industry Opportunities More than Private Firms? The Impact of Initial Firm Quality, WP

Private and public firms are potentially two very different animals



Getting there...



Baseline specification

$$\frac{I_{it}}{A_{it-1}} = \alpha\left(\frac{S_{it}-S_{it-1}}{S_{it-1}}\right) + \beta\left\{PUBLIC_i \times \left(\frac{S_{it}-S_{it-1}}{S_{it-1}}\right)\right\} + \delta\left(\frac{Z_{it}}{A_{it-1}}\right) + \phi\left\{PUBLIC_i \times \left(\frac{Z_{it}}{A_{it-1}}\right)\right\} + \mu_i + \eta_t + \varepsilon_{it}$$

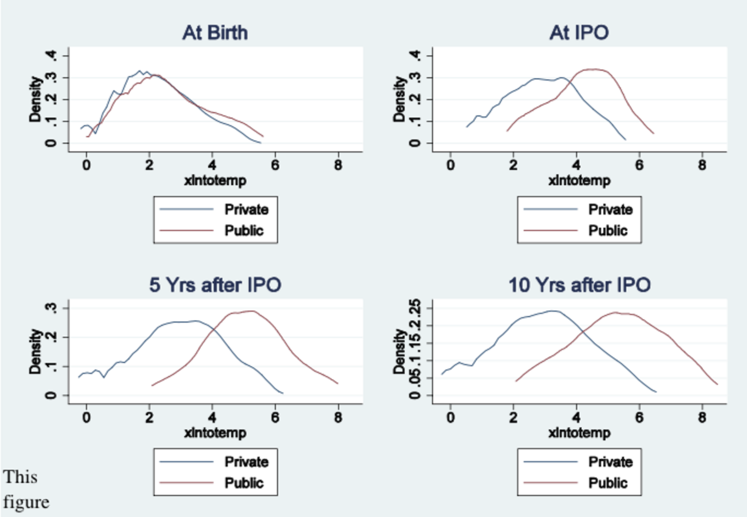
- S = sale
- Z = operating income
- Why *Z* × *Public*? Why not *Z* alone? Interpretation of single term *Z* here?
- Why Z and S alone but not Public?

Conclusion: listed frims react less to growth opportunities

Interpretation?

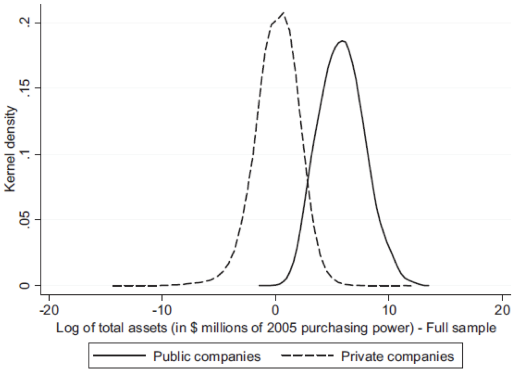
Proxy for inv. opps.	Sales growth			
Sample	Matched on size and industry (1)	Matched on size and industry (2)	Matched public firms only (3)	All public firms (4)
Investment opportunities	0.118***		0.028***	0.037***
	0.029		*0.006*	*0.005*
x public	-0.091***			
	0.030			
Inv. opps. (2002-2007)		0.180***		
		0.040		
x public		-0.143***		
		0.040		
Inv. opps. (2008-2011)		0.053***		
		0.019		
x public		-0.041**		
		0.021		

Maskimovic, Phillips, Yang (2023)



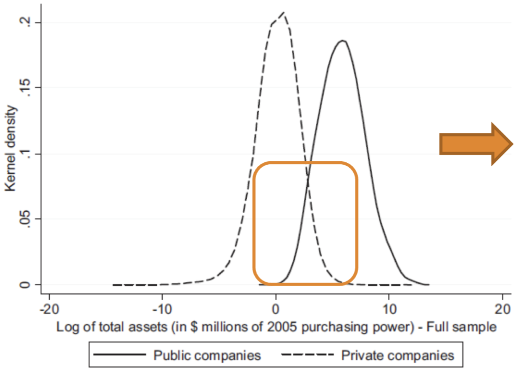
This figure

Unconditionally, public firms are bigger and more productive than private firms



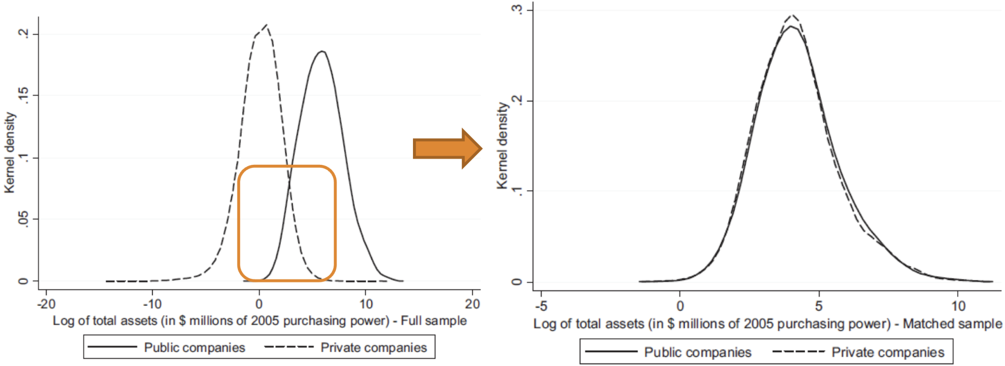
Public and private firms overlap only in a narrow size region

Unconditionally, public firms are bigger and more productive than private firms



Matching on size makes the two distributions coincide

Unconditionally, public firms are bigger and more productive than private firms



Comparison with AFL

- Noticeable facts?

	Total Emp	Wage (in $000)	Firm Age	Sales/Emp	Growth (last 3 years)	Growth (next 3 Years)
Public Unmatched	25703	45	21	10768	0.084	0.011
Public Matched	2479	62	17	2666	-0.104	-0.099
Private Matched	1343	43	17	1553	0.037	-0.008
Private Unmatched	22	29	12	135	0.000	0.000

Main takeaway

Controlling for ex-ante differences can generate more bias

Problem 2: DiD with only aggregate shifters (the dosage design)

- Common setting: you have an aggregate reform (e.g., reduction of labor cost for all unskilled workers)

- Can you estimate a DiD?

Problem 2: DiD with only aggregate shifters (the dosage design)

- Common setting: you have an aggregate reform (e.g., reduction of labor cost for all unskilled workers)
- Can you estimate a DiD?
- Standard solution: "dosage" treatment

$$Log(K_{i,t}) = \beta \frac{Unskilled_{i,t0}}{L_i} \times Post_t + FE$$

Problem 2: DiD with only aggregate shifters (the dosage design)

- Common setting: you have an aggregate reform (e.g., reduction of labor cost for all unskilled workers)
- Can you estimate a DiD?
- Standard solution: "dosage" treatment

$$Log(K_{i,t}) = \beta \frac{Unskilled_{i,t0}}{L_i} \times Post_t + FE$$

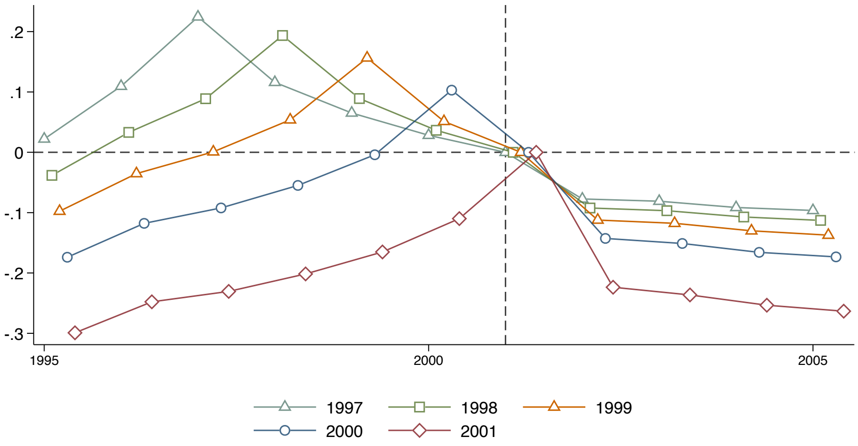
- Risk?

Problem 2: the exposure share peaks at the base year and mean-reverts

$$\frac{Unskilled_{i,t}}{L_{i,t}} = \beta \frac{Unskilled_{i,t0}}{L_i} \times Post_t + \alpha_i + \delta_t, \quad \text{where} t_0 \in [1997, .., 2001]$$

Problem 2: the exposure share peaks at the base year and mean-reverts

$$\frac{Unskilled_{i,t}}{L_{i,t}} = \beta \frac{Unskilled_{i,t0}}{L_i} \times Post_t + \alpha_i + \delta_t, \quad \text{where} t_0 \in [1997, .., 2001]$$



Side note: which variations should you use?

- Revenue = intermediate inputs + value-added (= wages + profits)

- Might be strange to use wage unskilled / total wages:

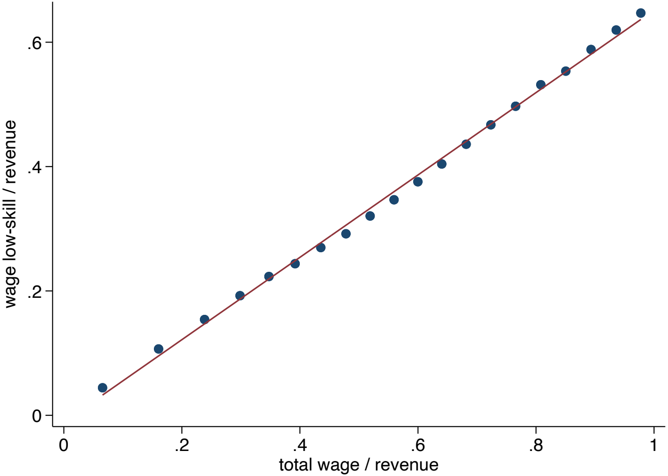
  - **Firm A**: one unskilled worker, but wage unskilled = 10% total cost $\rightarrow$ reduction of labor cost is a small shock but we assign a big one

  - **Firm B**: 5 unskilled workers, 3 skilled workers,but wage unskilled = 60% total cost $\rightarrow$ reduction of labor cost is a big shock but we assign a small one

$\rightarrow$ Solution?

Side note: which variations are you really using?

- Inverse is also potentially true: firms with high inputs and capital will always have a small shock no matter what



Side note: which variations are you really using?

- Inverse is also potentially true: firms with high inputs and capital will always have a small shock no matter what
- Other examples where this can be a problem:
  - Import competition shock / total workers or total output
    - Import only defined for manufacturing $\Rightarrow$ Import / total output $\approx$ share manufacturing
    - Ex: Autor et al. 2020
- In general: even with lots of variation on the positive support, if you have large share of zeros $\rightarrow$ potential pb

Closing: one decision rule

- Even with valid identification, your DiD coefficient is shaped by three choices:

  1. HDFE & representativeness: within-unit variation $\rightarrow$ special subsample, intensive margin

  2. Weights & granularity: implicit weights make firm / average / sum disagree and bias coefficients

  3. Functional form & aggregation: logs break additivity (Jensen) and drop the extensive margin

- Fix for problems 1–3: Disaggregation and aggregation invariance $\rightarrow$ consistent across levels, and lets you test SUTVA / spillovers (next lecture)

- Caveat: this does not fix mean reversion $\rightarrow$ use deep lags or non-mean-reverting covariates (industry, location, age)

Controlling for more, or disaggregating more, is not automatically cleaner

Thank you!

Adrien Matray adrien.matray@gmail.com